AS

AMNINDER SINGH

Post-Doctoral Scholar

CONTACT

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EDUCATION

Ph.D. | Environmental Sciences (Soil &Water)
UC Riverside
2018 – 2021

MS | Plant Science
California State University | Fresno
2015– 2017

BSc | Agriculture
Punjab Agricultural University, Ludhiana | Punjab, India
2011 – 2015

PROFESSIONAL EXPERIENCE

POST-DOCTORAL SCHOLAR
University of California, Riverside | Jan 2022 – Present

Focusing on the estimation of soil moisture by combining public and private soil moisture ground measurements, remote sensing, and other datasets with machine learning methods.

GRADUATE STUDENT RESEARCHER
University of California, Riverside | Jan 2018 – Dec 2021

Dissertation: Advancing Urban Landscape Irrigation Management using Smart Controllers and Machine Learning-Based Models

- Experienced in sampling and analyzing soil hydrology field measurements, including, soil water potential, soil volumetric water content, infiltration, evapotranspiration, and meteorological factors.
- Aggregating & statistically analyzing large agricultural datasets using Python, R or MATLAB.

CO-PRESIDENT, MINI-GSA
Department of Environmental Sciences | UCR | 2020 – 2021

TEACHING ASSISTANT, INTRODUCTION TO ENVIRONMENTAL SCIENCE: NATURAL RESOURCES (ENSC001)
Department of Environmental Sciences | UCR | Fall 2020

GRADUATE RESEARCH ASSISTANT
CSU, Fresno | Aug 2015 – Dec 2018

Thesis: Use of EM-38 soil surveys in forage fields at a saline drainage water reuse site to calibrate a hydro-salinity model for decision support.

- Experience conducting soil salinity mapping using electromagnetic sensing platforms.

GRADUATE TEACHING ASSISTANT, SWL100- (Soils Lab)
CSU, Fresno | Fall 2015 & Fall 2016

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SKILLS

- Statistical and programming skills: Python, Matlab, and R
- Machine Learning
- ArcGIS
- Soil and environmental sensor network data acquisition systems including data acquisition hardware and software.
- Soil sensing technologies.
- SQL
- HTML & CSS

LANGUAGES

- English
- Punjabi
- Hindi

AFFILIATIONS

- Soil Science Society of America (SSSA)
- American Society of Agronomy (ASA).
- Crop Science Society of America (CSSA).
- American Society of Agricultural and Biological Engineers (ASABE).

ACHIEVEMENTS AND AWARDS

- Best Oral Presentation Award - 2021 UCR Environmental Sciences Graduate Student Symposium.
- Dissertation Year Program Fellowship, Department of Environmental Sciences, UCR (Fall 2021).
- Hilda and George Liebig Environmental Sciences Summer Fellowship, Department of Environmental Sciences, UCR (2020).
- SCSC-NWRI Fellowship, Southern California Salinity Coalition (SCSC) and the National Water Research Institute (NWRI), 2018-2020.
- Stolzy-Letey Travel Scholarship, Department of Environmental Sciences, UCR (2018-2019).
- Mini GSA Award, Hilda and George Liebig Foundation, Department of Environmental Sciences, UCR (2019, 2021).
- UCANR Drone Camp, June 18-20, 2018, UC San Diego
- Outstanding Thesis Award. Jordan College of Agricultural Sciences & Technology, CSU, Fresno (2017-2018 academic year).
- Graduate Student Achiever, Department of Plant Science, CSU, Fresno. (2018).
- Ag One-John P. “Phil” Larson Scholarship, CSU, Fresno (Spring 2017).
- Inducted to The Honor Society of Phi Kappa Phi (Spring 2017).
- Jim Pattersen’s Ag Science student recognition award - Agriculture Science Student of the Year, 23rd Assembly District, (2017).
- The Gerald O. Mott Meritorious Graduate Student Award in Crop Science, Crop Science Society of America (CSSA), (2017).
- Third place in Poster competition, Plant and soil conference – American Society of Agronomy, California chapter, (2017).
- Jordan Assistantship, Jordan College of Agricultural Sciences and Technology, CSU, Fresno (August 2015-May 2017).
- Graduate Research Fellowship, Graduate Net Initiative, CSU, Fresno (2016-2017).
- Travel grant, Division of Graduate Studies, CSU, Fresno for the 2016 ASA, CSSA, and SSSA Annual Meeting, November 5-9, 2016, in Phoenix, AZ.
- Dean’s Scholarship Tuition Waiver, CSU, Fresno (2015- 2016).
- Punjab State Marketing Board Scholarship, PAU, Ludhiana (2011 - 2015).

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PRESENTATIONS

- **Singh A**., *Haghverdi A., Sapkota A., Haver D*., Recycled Water Application for Turfgrass Irrigation to conserve water and maintain turf health, **ASABE 2021 Annual International Meeting** | July 12–16, 2021 | Virtual (Poster presentation).
- ***Singh A**., Haghverdi A., Sapkota A., Haver D.,* Autonomous Irrigation in Bermudagrass Using Recycled Water and Soil Moisture Sensor Based Smart Controller, **2020 ASA-CSSA-SSSA International Annual Meeting** | November | Virtual (Oral presentation)
- ***Singh A**., Haghverdi A. Öztürk H. S., Durner W.,* Pseudo Continuous Water Retention Pedotransfer Functions for International Soils Measured with the Evaporation Method and the HYPROP System, **2020 ASA-CSSA-SSSA International Annual Meeting** | November | Virtual (Poster)
- ***Singh A**., Haghverdi A., Sapkota A., Haver D.,* Autonomous Bermudagrass irrigation using recycled water and SMS based smart controller, **UC Riverside Urban landscape Irrigation and Water Conservation Virtual Field Day for Master Gardeners** | September 3, 2020.
- ***Singh A**., Sharifi M., Ghodsi S., Haghverdi A.,* Monitoring the impact of multiple water conservation strategies on turfgrass using ground based remote sensing tools. **2019 SSSA Annual Meeting, San Diego, CA** *(Oral presentation).*
- *Benes S.E., **Singh A**., Quinn N.W. and Cassel Sharma F.* (2018). Monitoring Soil Salinity using EM-38 Surveys to Calibrate a Hydro-salinity Model for Decision Support. 2018, The Future of Water for Irrigation in California and Israel Workshop. Davis, California, July 16-18, 2018. *(Poster)*
- ***Singh, A**., Benes, S.E., Quinn, N., Cassel, F.* Use of EM-38 Soil Surveys in Forage Fields at a Saline Drainage Water Reuse Site to Calibrate a Hydro-salinity Model for Decision Support. **Plant and Soil Conference by American Society of Agronomy – California Chapter** (2018) *(Poster).*
- ***Singh, A**., Benes, S.E., Quinn, N., Cassel, F.* Monitoring Soil Salinity in Alfalfa and 'Jose' tall wheatgrass fields using EM-38 soil Surveys and Developing Input Data for a Transient Hydro-salinity Computer Model. **38th Annual Central California Research Symposium**, University Business Center, CSU, Fresno, April 18, 2017 *(Oral presentation).*
- ***Singh, A**., Benes, S.E., Quinn, N., Cassel, F.* Use of EM-38 soil salinity surveys to develop validation data sets for a transient hydro-salinity model CSUID-II. **CWEMF annual meeting**, Folsom, CA, March 20-22, 2017 *(Poster).*
- ***Singh, A**., Benes, S.E., Quinn, N., Cassel, F.* Development of validation data sets for a transient hydro-salinity model using EM-38 soil surveys, irrigation water monitoring and forage analysis. **Plant and Soil Conference by American Society of Agronomy – California Chapter**, Jan. 31- Feb. 1, 2017 *(Poster).*
- ***Singh, A**., Benes, S.E., Quinn, N., Cassel, F., Bottino, U. Jr.,* Soil Salinity Mapping Using Electromagnetic Induction (EM-38) to Provide Input Data for the CSUID-II Model, a Decision Support Tool for Irrigation in Saline Water Reuse Areas. **2016 ASA, CSSA, and SSSA Annual Meeting**, Phoenix, AZ, November 6-9, 2016 *(Poster).*

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PUBLICATIONS (peer-reviewed)

- Haghverdi, A., M. Reiter, **A. Singh**, and A. Sapkota. 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: II. Assessment of NDVI, CWSI, and Canopy Temperature Dynamics. Agronomy 11(9): 1733 Available at https://www.mdpi.com/2073-4395/11/9/1733.
- Haghverdi, A., M. Reiter, A. Sapkota, and **A. Singh**. 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: I. Assessment of Visual Quality, Soil Moisture and Performance of an ET-Based Smart Controller. Agronomy 11(8): 1666.
- **Singh, A**., A. Haghverdi, H.S. Öztürk, and W. Durner. 2021. Developing Pseudo Continuous Pedotransfer Functions for International Soils Measured with the Evaporation Method and the HYPROP System: II. The Soil Hydraulic Conductivity Curve. Water 13(6): 878 Available at https://www.mdpi.com/2073-4441/13/6/878.
- Haghverdi, A., **A. Singh**, A. Sapkota, M. Reiter, and S. Ghodsi. 2021. Developing irrigation water conservation strategies for hybrid bermudagrass using an evapotranspiration-based smart irrigation controller in inland southern California. Agric. Water Manag. 245(April 2020): 106586 Available at https://doi.org/10.1016/j.agwat.2020.106586.
- **Singh, A**., A. Haghverdi, H.S. Öztürk, and W. Durner. 2020. Developing pseudo continuous pedotransfer functions for international soils measured with the evaporation method and the hyprop system: I. the soil water retention curve. Water (Switzerland) 12(12): 1–17 Available at https://www.mdpi.com/2073-4441/12/12/3425.
- **Singh, A**., N.W.T. Quinn, S.E. Benes, and F. Cassel. 2020. Policy-Driven Sustainable Saline Drainage Disposal and Forage Production in the Western San Joaquin Valley of California. Sustain. 2020, Vol. 12, Page 6362 12(16): 6362 Available at https://www.mdpi.com/2071-1050/12/16/6362/htm
- **Singh, A**., A. Haghverdi, M. Nemati, and J. Hartin. 2020. Efficient Urban Water Management Smart Weather-Based Irrigation Controllers. Univ. Calif. Agric. Nat. Resour. 8674(July): 1–11